Lecture 4: Power Diodes

EEE 339 Power Electronics · Complete study notes with previous-year solutions · Source: *class slides/Lecture 04 Power Diodes.pdf* (Rashid ch. 2)

Exam priority (from the 2020 to 2024 papers):

1. **Reverse recovery**: definition, waveform, significance. Asked in TT-1 2020, 2021 Q2a, 2023 Q1b.
2. **Reverse recovery numerical** (di/dt = 20 A/µs, t_rr = 5 µs). Asked in **2021 Q4c and 2023 Q1c with the same numbers**.
3. Diode types (general-purpose / fast-recovery / Schottky / SiC), soft vs hard recovery.
4. Series and parallel diodes; freewheeling diode (2022 Q1a short note).

Expected weight in your 60-mark paper: one 3 to 6 mark sub-part in Part A.

1. Dynamic (switching) characteristics

When a diode switches, its current and voltage do not change instantly. The slide divides the transition into intervals:

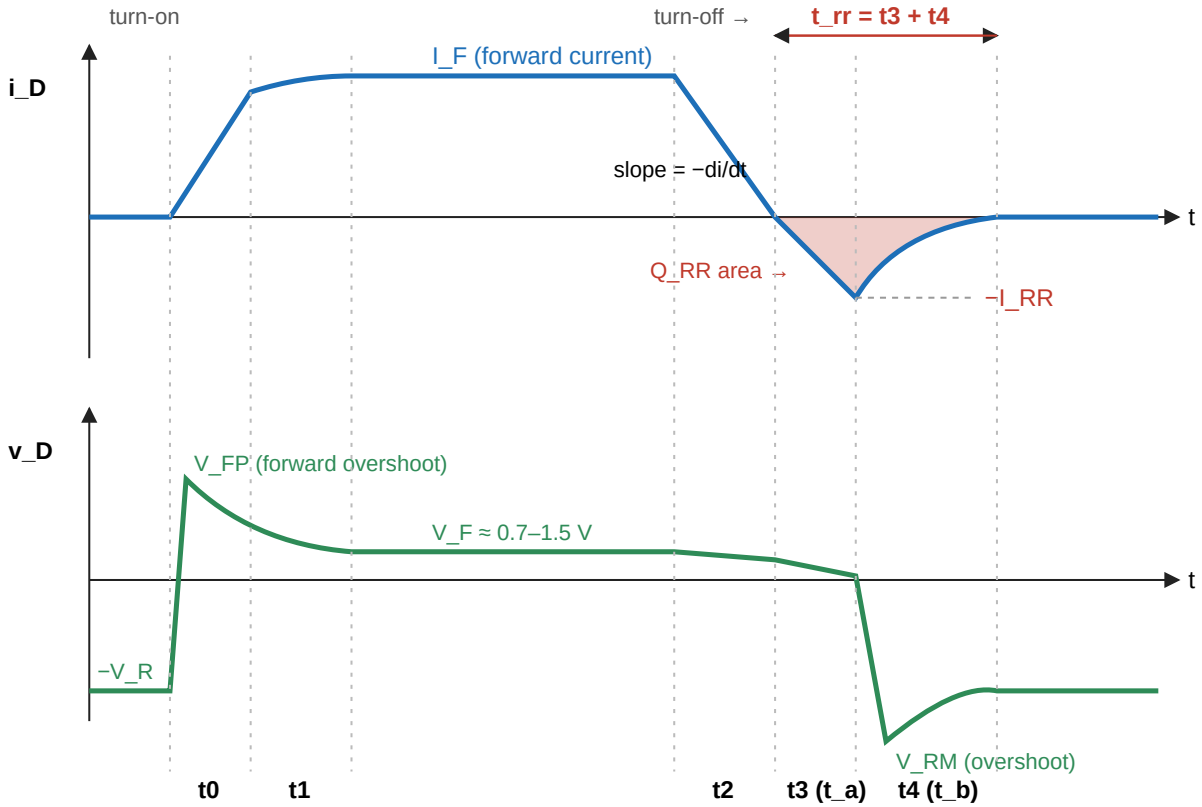


Fig. 1: Diode turn-on (t0, t1) and turn-off (t2, t3, t4). Shaded area = reverse recovery charge Q_RR.

Interval	What happens
t0 (turn-on)	Source goes from negative to positive. Stored charge in the depletion region is removed; the current starts to rise. The diode voltage shows a **peak (forward overshoot V_FP)** because carrier injection is not yet complete and the N-region resistance is high.
t1	Carrier injection completes and the voltage falls to the forward drop V_F. t0 and t1 are tens to hundreds of ns in power diodes. End of t1 = turn-on complete.
t2 (turn-off starts)	Source goes from positive to negative. The **minority carriers** (electrons in P, holes in N) must be removed before the potential barrier forms, so the diode **keeps conducting**. The current falls at the rate di/dt; the voltage falls slightly as R_ON falls.
t3 (= t_a)	The current goes **negative** and reaches its negative peak **−I_RR**. At this point the stored minority carriers have been swept out and the diode starts to become reverse-biased.
t4 (= t_b)	The current returns to zero as the depletion region recharges. The diode voltage shows a **reverse overshoot** because of the circuit's parasitic (stray) inductance.
t_rr	**Reverse recovery time t_rr = t3 + t4** (t_a + t_b). The area under the time axis in the current waveform is the **reverse recovery charge Q_RR**.

2. Reverse recovery (most important topic)

Definitions

- **Reverse recovery time (t_rr):** the time from the instant the forward current crosses zero until the reverse current has decayed to about 25% of its peak I_RR (practically, back to zero); t_rr = t_a + t_b.
- **Peak reverse recovery current (I_RR):** the maximum reverse current that flows during recovery, at the end of t_a.
- **Reverse recovery charge (Q_RR):** the charge carried by the reverse current during recovery, equal to the area of the reverse-current triangle (shaded in Fig. 1).
- **t_a:** time to remove stored charge from the depletion region. **t_b:** time to remove stored charge from the bulk semiconductor.
- **Softness factor S = t_b / t_a.**

Formulas (and where they come from)

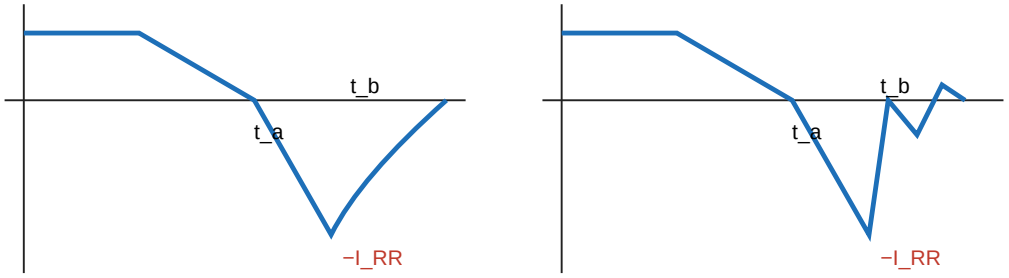
> Approximate the reverse current as a triangle with height I_RR and base t_rr:
> $Q_{RR} \approx \frac{1}{2} \cdot I_{RR} \cdot t_{rr}$ ……(1)
> The current falls at a constant rate di/dt during t_a:
> $I_{RR} = t_a \cdot (di/dt)$ ……(2)
> For an abrupt recovery (t_b small), t_a ≈ t_rr, so I_RR ≈ t_rr · (di/dt). Substituting into (1):
> $Q_{RR} = \frac{1}{2} \cdot (di/dt) \cdot t_{rr}^2$ ……(3)
> Rearranging (3) and combining with (2):
> $t_{rr} = \sqrt{(2\ Q_{RR} / (di/dt))}$ ……(4) $I_{RR} = \sqrt{(2\ Q_{RR} \cdot (di/dt))}$ ……(5)
> **Remember:** t_rr and I_RR depend on the stored charge Q_RR and on di/dt (which the circuit sets). Q_RR grows with the forward current I_F.

Significance (importance) of t_rr

1. **Sets the maximum switching frequency.** Reverse recovery must finish before the diode is forward-biased again. The lower t_rr, the faster the diode can be switched (fast / ultrafast diodes).
2. **Switching losses.** During t_rr the diode carries reverse current while voltage builds up across it, so energy is lost each cycle. Loss rises with frequency.

3. **Stress on the other switch.** I_RR adds to the current of the transistor that turns on, increasing its peak current and turn-on loss.
4. **Overvoltage and EMI.** A fast fall of reverse current through stray inductance gives v = L·di/dt spikes, overvoltage and electromagnetic interference.
5. **Choice of diode.** Rectifiers at line frequency can use general-purpose diodes; choppers and inverters need fast-recovery or Schottky diodes.

Soft vs hard recovery



Soft recovery (t_b large, S > 1) **Hard / abrupt recovery (t_b small, S < 1)**

Fig. 2: Shape of the current after −I_RR.

	Soft recovery	**Hard (abrupt) recovery**
Current fall after I_RR	Slow	Fast (snappy)
Softness S = t_b/t_a	Large (S ≈ 1 or more)	Small (S < 1)
Voltage spike L·di/dt	Low	High, causes oscillation and EMI
Preferred?	**Yes** (the slide: "a slow current fall is more attractive")	No

3. Types of diodes

Common types (slide 6)

Type	**Key points**
Small-signal diode	Low voltage, low current; low-power rectifiers, limiters, amplifiers, waveshaping. Designer must know V_F, breakdown voltage, I_RR and t_rr.
Power rectifier diode	High forward current and high reverse breakdown voltage; rectifiers, inverters, power supplies. Thermal resistance matters; usually needs a heat sink.
Zener diode	Voltage regulation; operates in reverse breakdown; the Zener current clamps the voltage.
Schottky diode	Thin metal film on the semiconductor gives a rectifying contact. V_F ≈ 0.3 V; low-voltage applications.

Power diode categories (by recovery): learn this table

	General-purpose (standard)	Fast-recovery	Schottky
t_rr	High, typically **25 µs**	Low, **< 5 µs**	Almost none (only junction capacitance)
Voltage rating	50 V to about **5 kV**	50 V to about **3 kV**	Low (about 100 V); V_F ≈ 0.3 V
Current rating	< 1 A to several thousand A	< 1 A to hundreds of A	Low to medium
Carriers	Minority + majority (PN)	Minority + majority (PN, lifetime-controlled)	**Majority carriers only**: no excess minority carriers to recombine
Use	Low-speed: line-frequency rectifiers	DC-DC and DC-AC converters	High-frequency, low-voltage supplies

Why Schottky recovers so fast: conduction is by majority carriers only, so there is no stored minority charge. The small recovered charge comes from the junction capacitance only, so it is **largely independent of the reverse di/dt**.

Silicon carbide (SiC) diodes

- SiC outperforms Si and GaAs. SiC Schottky diodes have **no reverse recovery time**, ultrafast switching, and switching that is **not affected by temperature**.
- Benefits: lowest switching losses (low Q_RR), surge-current stable and rugged, lower cooling cost, higher frequency and power density, low device capacitance, less EMI.
- Uses: power supplies, solar inverters, transport, welding, air conditioners.
- A Schottky diode is made by joining doped n-type semiconductor with a metal such as gold, silver or platinum.

4. Series and parallel connection

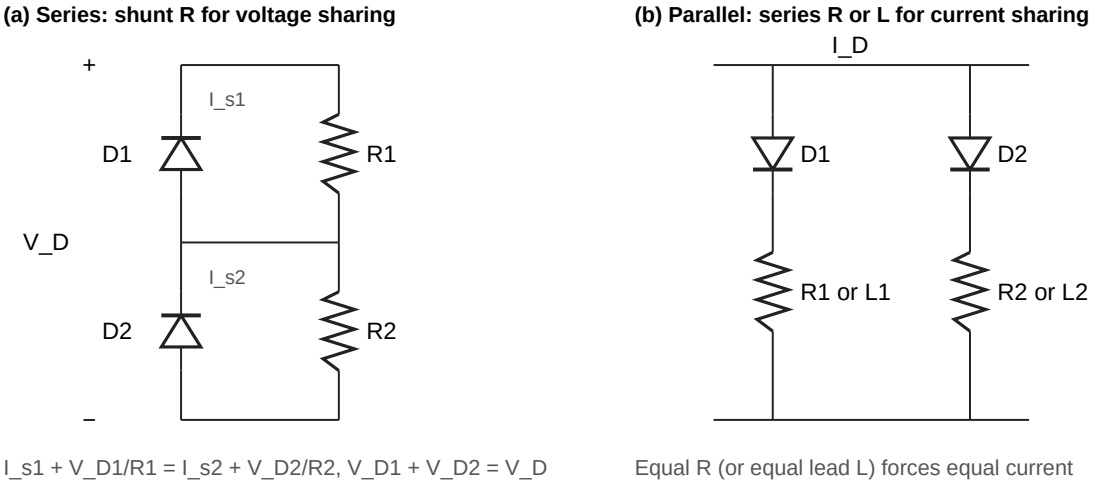


Fig. 3: (a) Series diodes with voltage-sharing resistors. (b) Parallel diodes with current-sharing resistors (or equal inductances).

Series-connected diodes (for high voltage, e.g. HVDC)

- **Why:** one diode cannot block the required voltage.
- **Problem:** diodes of the same type have different v-i curves (production tolerance). In reverse blocking they all carry the **same leakage current**, so their **blocking voltages can be very different**; one diode may break down.

- **Solution:** connect a **resistor across each diode** to force equal voltage sharing. The leakage currents of the diodes then differ, but each diode-plus-resistor pair carries the same total current. (Capacitors are added across each diode to share voltage during transients.)
- Equations: $I_{s1} + V_{D1}/R1 = I_{s2} + V_{D2}/R2$ and $V_{D1} + V_{D2} = V_D$.

Parallel-connected diodes (for high current)

- **Why:** one diode cannot carry the required current.
- **Problem:** current divides according to the forward voltage drops; the diode with the lowest V_F hogs the current.
- **Solution:** equal series **inductances** (e.g. in the leads) or **current-sharing resistors** (the resistors waste power, so they may not be practical). Choosing diodes of the same type with equal V_F also helps.
- In parallel, every diode sees the **same reverse voltage**.

5. Applications of diodes

- **Rectifiers** (AC to DC) and **voltage multipliers**.
- **Freewheeling diode:** gives the inductor current a path when the transistor turns off.

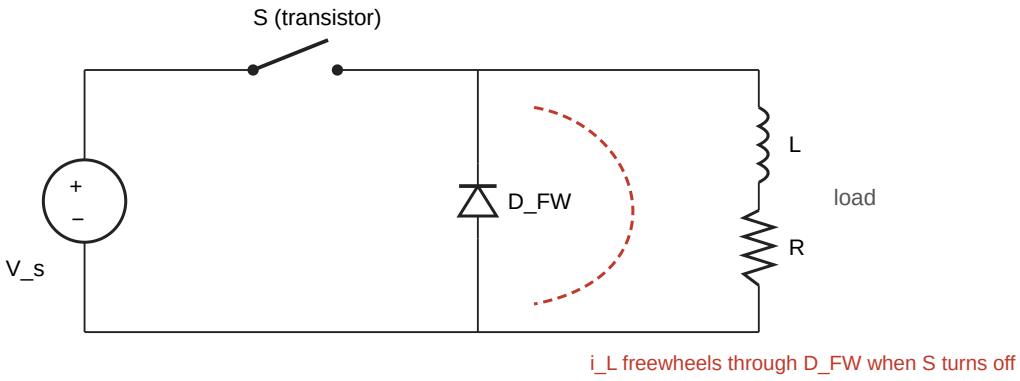


Fig. 4: Freewheeling diode D_FW across an inductive load.

Without D_FW, turning the switch off would force the inductor current to zero instantly. That gives a huge $v = L \cdot di/dt$ across the switch, which would destroy it. With D_FW, the inductor voltage reverses and forward-biases the diode, so the current continues smoothly through D_FW and the switch is protected.

6. Previous-year questions with full solutions

TT-1 2020 **Q. Explain the reverse recovery time of a diode with diagram. [4]**

Answer. Draw Fig. 1 (current part is enough, label I_F, −I_RR, t_a, t_b, t_rr, Q_RR).
When a conducting diode is suddenly reverse-biased, the current does not stop at zero. The stored minority carriers in the junction and bulk must first be removed, so the current falls at the rate di/dt, passes through zero and flows in the reverse direction.

- During **t_a** the charge in the depletion region is removed and the reverse current rises to its peak **I_RR**.
- During **t_b** the charge in the bulk semiconductor is removed and the reverse current decays to zero.
- **t_rr = t_a + t_b** is the reverse recovery time; the area of the reverse current is the reverse recovery charge **Q_RR ≈ ½ I_RR t_rr**.

The diode can block reverse voltage only after t_rr has ended.

2021 Q2(a) **Q. Discuss the significance of reverse recovery time of a diode. [5]**

Answer. Start with a one-line definition plus the current sketch (Fig. 1), then give the points:

1. **Maximum switching frequency:** recovery must be over before the diode is forward-biased again, so t_rr limits how fast the diode can switch. Low t_rr means fast/ultrafast diodes.
2. **Switching loss:** reverse current flows while reverse voltage builds up, so energy is lost every cycle; the loss is proportional to switching frequency.
3. **Extra stress on the complementary switch:** the I_RR adds to the transistor current at turn-on (higher peak current, higher loss).
4. **Voltage spikes and EMI:** the rapid fall of reverse current through stray inductance gives L·di/dt overvoltage and noise. Soft-recovery diodes reduce this.
5. **Diode selection:** line-frequency rectifiers use general-purpose diodes (t_rr ≈ 25 µs); choppers and inverters need fast-recovery (< 5 µs) or Schottky diodes.

2023 Q1(b) **Q. Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters? [6]**

Answer.

- **Reverse recovery time t_rr:** the time between the instant the forward current falls to zero and the instant the reverse current decays back to (almost) zero; t_rr = t_a + t_b.
- **Reverse recovery current I_RR:** the peak reverse current that flows during recovery, at the end of t_a; I_RR = t_a·(di/dt) ≈ √(2Q_RR·di/dt).

Draw Fig. 1 (current waveform) with t_a, t_b, t_rr, I_RR and Q_RR marked.

Importance:

1. t_rr fixes the highest usable switching frequency.
2. I_RR and t_rr together set the reverse recovery loss (∝ Q_RR · V_R · f).
3. I_RR flows through the switch that turns on, increasing its peak current and stress.
4. The fall of I_RR through stray inductance causes overvoltage and EMI, which is why soft recovery is preferred.
5. They decide which diode type to use (general-purpose, fast-recovery or Schottky).

2021 Q4(c) [4] 2023 Q1(c) [6] **Q. The manufacturer of a selected diode gives the rate of fall of the diode current di/dt = 20 A/μs, and its reverse recovery time t_rr = 5 μs. What value of peak reverse current (and storage charge) do you expect?**

Given: di/dt = 20 A/μs = 20 × 10^6 A/s, t_rr = 5 μs = 5 × 10^{-6} s.
Step 1: stored charge. Q_RR = ½ (di/dt) t_{rr}^2 = ½ × 20 × 10^6 × (5 × 10^{-6})² = ½ × 20 × 10^6 × 25 × 10^{-12} = **250 μC**
Step 2: peak reverse current. I_RR = √(2 Q_RR (di/dt)) = √(2 × 250 × 10^{-6} × 20 × 10^6) = √10000 = **100 A**
Check: I_RR = t_rr × di/dt = 5 μs × 20 A/μs = 100 A ✓

Answer: I_RR = 100 A, Q_RR = 250 μC

Assumption to write in the exam: abrupt recovery (t_b small), so t_a ≈ t_rr.

2022 Q1(a)(ii) **Q. Write a short note on: Freewheeling diode. [about 4]**

Answer. A freewheeling (flywheel) diode is connected across an inductive load (anti-parallel to the load), as in Fig. 4.

- **Purpose:** when the switch (or the rectifier diode/SCR) turns off, the inductor current cannot stop instantly. The freewheeling diode gives it a closed path, so the current continues smoothly.
- **Protects the switch:** without it, the inductor would create a very large L·di/dt voltage across the switch and destroy it.
- **In rectifiers** (e.g. half-wave with RL load): it stops the output voltage going negative, so V_o increases (V_o = V_m/π), the load current becomes smoother (continuous for large L), and the input power factor improves. The rectifier diode no longer conducts in the negative half cycle.
- Used in choppers (DC-DC converters), rectifiers with inductive loads, relay and motor drives.

7. Practice questions (likely next) with short answers

Predicted **Q. A diode has Q_RR = 1500 µC and di/dt = 30 A/µs. Find t_rr and I_RR.**

t_rr = √(2Q_RR/(di/dt)) = √(2 × 1500 × 10^{-6} / 30 × 10^{6}) = **10 µs**; I_RR = √(2Q_RR·di/dt) = √(2 × 1500 × 10^{-6} × 30 × 10^{6}) = **300 A** (check: 10 µs × 30 A/µs = 300 A ✓)

Predicted **Q. Two diodes are connected in series to block V_D = 5 kV. Leakage currents I_s1 = 30 mA, I_s2 = 35 mA. (a) Find each diode voltage if R1 = R2 = 100 kΩ. (b) Find R1 for equal sharing if R2 = 100 kΩ.**

(a) I_s1 + V_D1/R = I_s2 + V_D2/R with V_D2 = V_D − V_D1 ⇒ V_D1 = V_D/2 + R(I_s2 − I_s1)/2 = 2500 + 100 kΩ × 5 mA / 2 = **2750 V**, V_D2 = **2250 V**.
(b) V_D1 = V_D2 = 2500 V: I_s1 + 2500/R1 = I_s2 + 2500/R2 ⇒ 2500/R1 = 35 mA − 30 mA + 25 mA = 30 mA ⇒ **R1 = 83.33 kΩ**.

This textbook example (Rashid) is not on your slides, but the slide explains the method, so an examiner can ask it.

Predicted **Q. Compare general-purpose, fast-recovery and Schottky diodes. / Why does a Schottky diode have almost no reverse recovery?**

Use the table in Section 3. Schottky: majority-carrier device, so no stored minority charge; only junction capacitance gives a small recovered charge, independent of di/dt; low V_F (≈ 0.3 V) but low reverse voltage rating.

Predicted **Q. What are soft and hard recovery? Which is preferred and why?**

Soft recovery: slow fall of reverse current (large t_b, S = t_b/t_a large). Hard recovery: fast, snappy fall. Soft is preferred because the voltage induced in stray inductances (L·di/dt) is lower, so less overvoltage, oscillation and EMI.

Predicted **Q. Why are resistors connected across series diodes and resistors/inductors in series with parallel diodes?**

Series: diodes of the same type have different leakage characteristics, so with the same leakage current their reverse voltages differ; shunt resistors force equal voltage sharing.
Parallel: current divides according to V_F, so the diode with the lowest V_F carries most current; equal series resistors or lead inductances force equal current sharing.

8. One-look revision sheet

Item	Remember
Turn-on intervals	t0: current rises, forward overshoot V_FP; t1: falls to V_F
Turn-off intervals	t2: still conducting, current falls; t3 (t_a): reverse current to −I_RR; t4 (t_b): back to 0, reverse overshoot from stray L
t_rr, softness	t_rr = t_a + t_b = t3 + t4; S = t_b/t_a
Formulas	$Q_{RR} = \frac{1}{2} I_{RR} t_{rr} \cdot I_{RR} = t_a \, di/dt \cdot t_{rr} = \sqrt{(2Q_{RR}/(di/dt))} \cdot I_{RR} = \sqrt{(2Q_{RR} \, di/dt)}$
Classic numerical	20 A/µs, 5 µs gives **I_RR = 100 A, Q_RR = 250 µC**
Diode types	General-purpose 25 µs, up to 5 kV · Fast-recovery < 5 µs, up to 3 kV · Schottky: majority carriers, V_F ≈ 0.3 V, low voltage
SiC Schottky	No t_rr, ultrafast, temperature-independent switching, low loss, less EMI
Series / parallel	Series: shunt R (voltage sharing) · Parallel: series R or equal L (current sharing)
Freewheeling diode	Path for inductor current at switch-off; prevents L·di/dt spike; in rectifiers keeps v_o ≥ 0

Numbers in the solutions are computed by Exam_Prep/src/lec04_notes.py.